

Vegetation Change & Climate Stress in the Sierra Nevada



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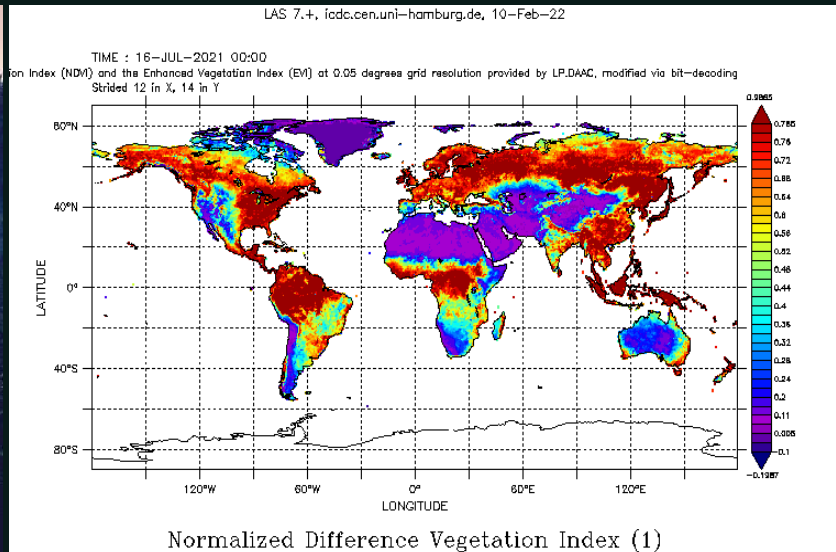
Why Study Vegetation Change?

- Forests act as major carbon sinks storing CO₂ in biomass
- Drought, extreme heat, and wildfires reduce recovery capacity
- NDVI = measure of vegetation greenness and photosynthetic activity
- Objectives (2024 focus):
 - Quantify vegetation health using MODIS NDVI
 - Analyze seasonal NDVI trends and temporal patterns
 - Compare NDVI with temperature & precipitation from Cal-Adapt



MODIS NDVI and GIS Processing Workflow

- Dataset: MODIS MOD13Q1 (250 m, 16-day NDVI)
- Year 2024 tiles covering Sierra Nevada downloaded from NASA EarthData
- Pre-processing in QGIS: clip to Sierra boundary & reproject (EPSG 3310)
- Consistent NDVI color ramp for visual comparison
- Exported GeoTIFFs for Python analysis



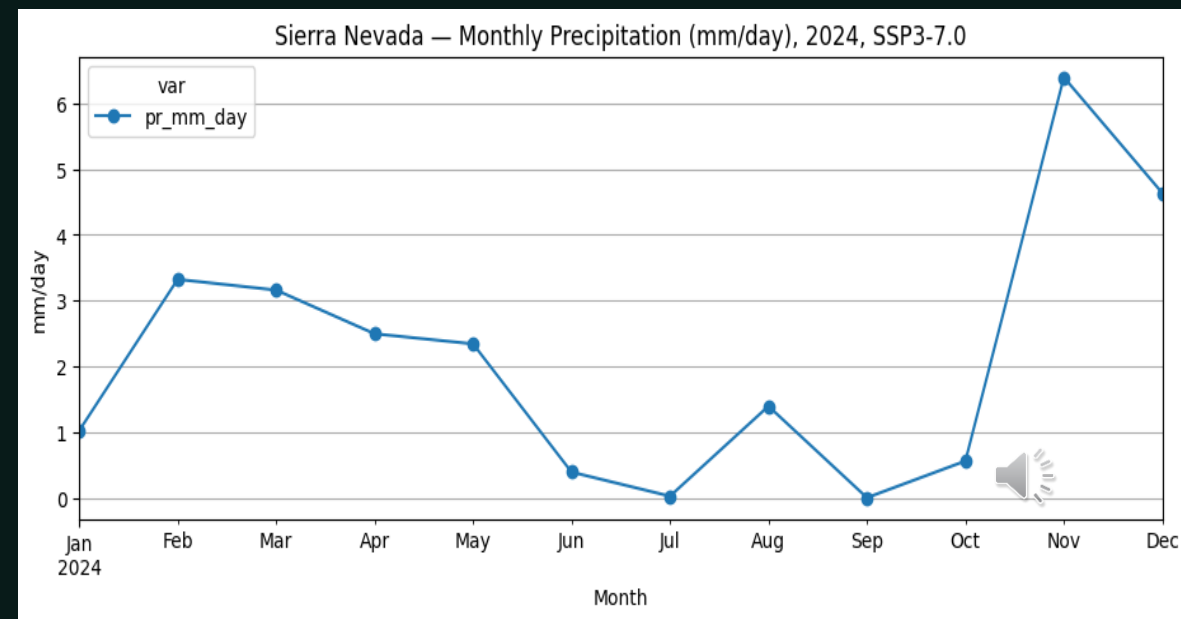
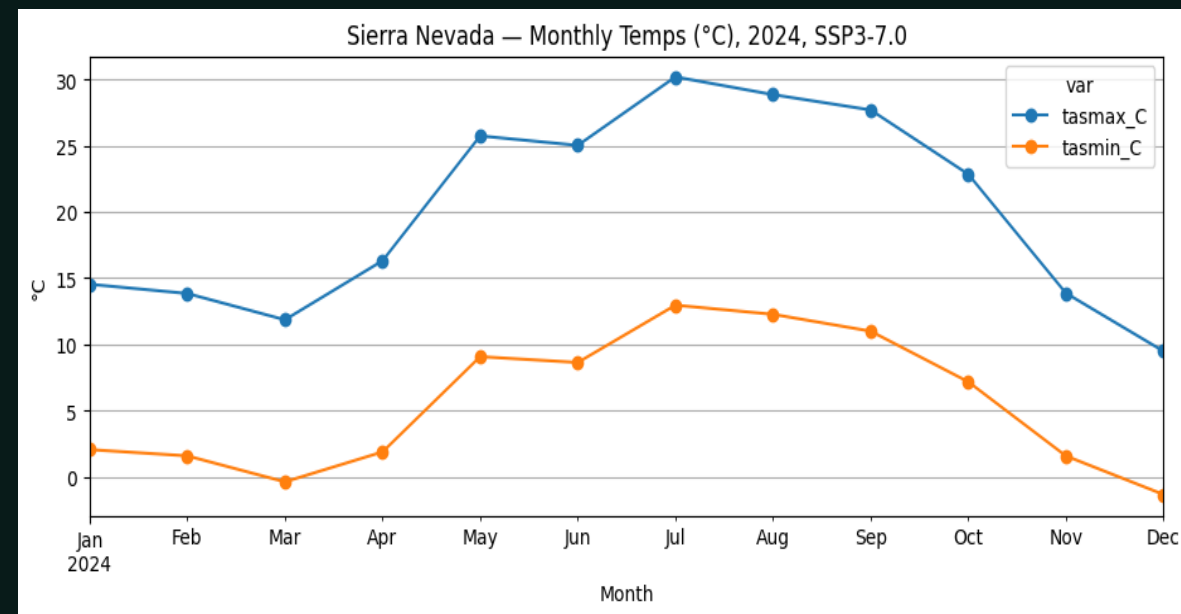
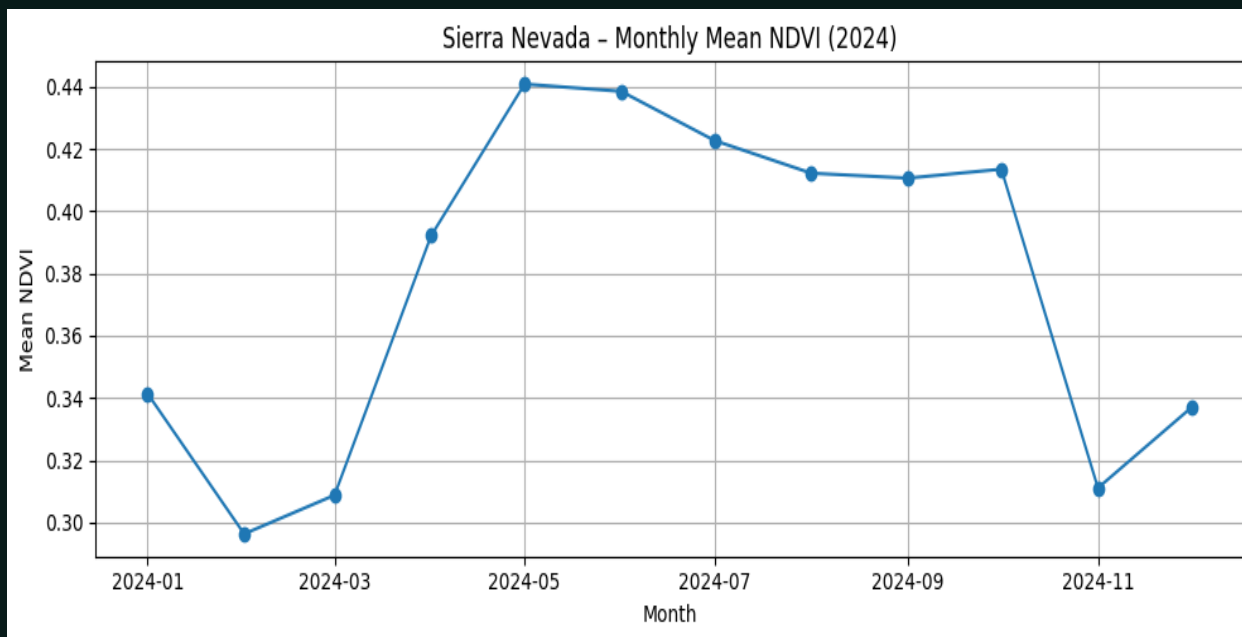


Temporal and Climate Analysis in Python

- Libraries: rioxarray, pandas, matplotlib
- Converted NDVI to reflectance (-1 to $+1$ scale)
- Computed mean NDVI per 16-day composite
- Cal-Adapt (SSP3-7.0 ACCESS-CM2) for temperature & precipitation
- Regression + K-Means clustering to find NDVI–climate relationships

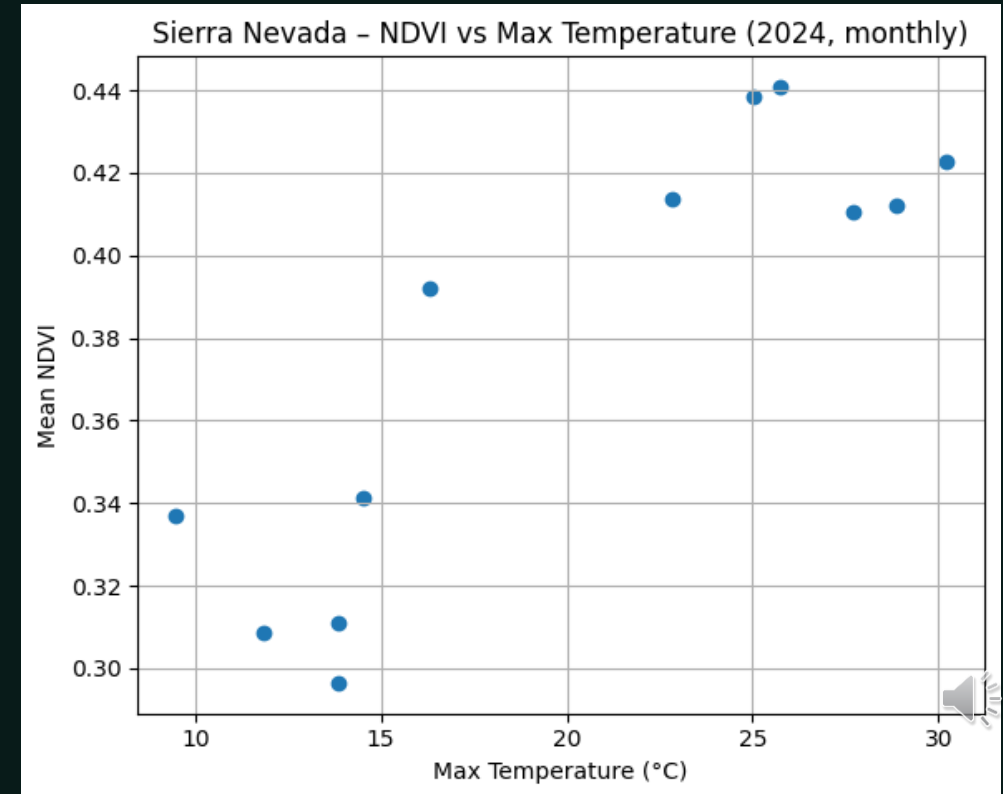
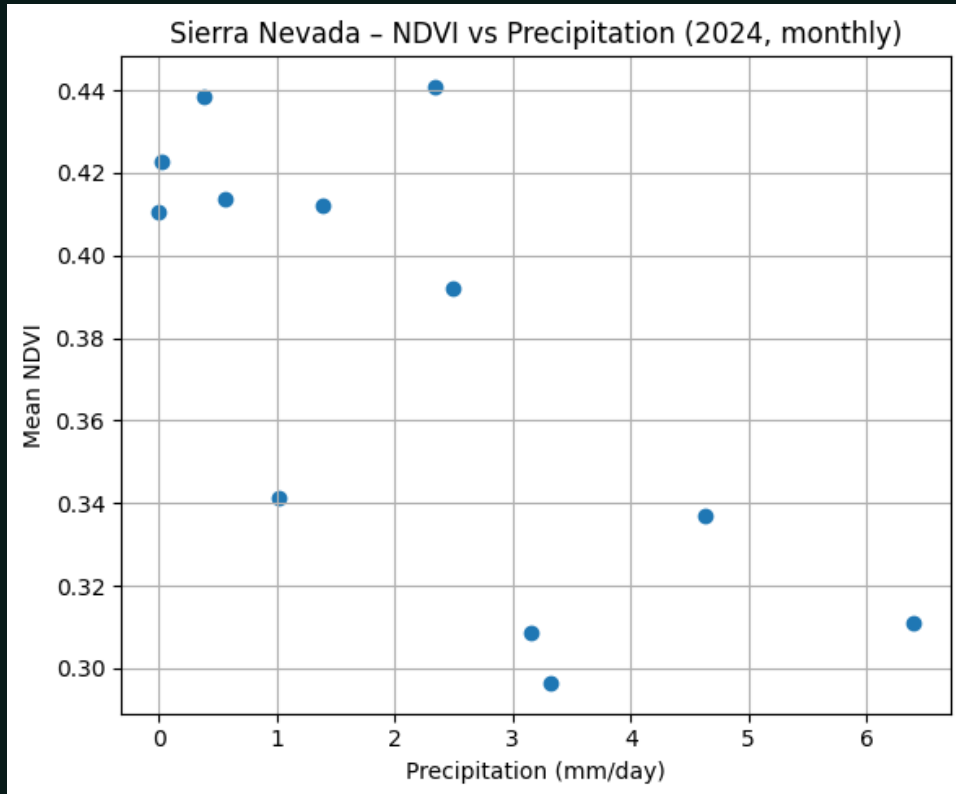


Seasonal NDVI and Climate Patterns (2024)





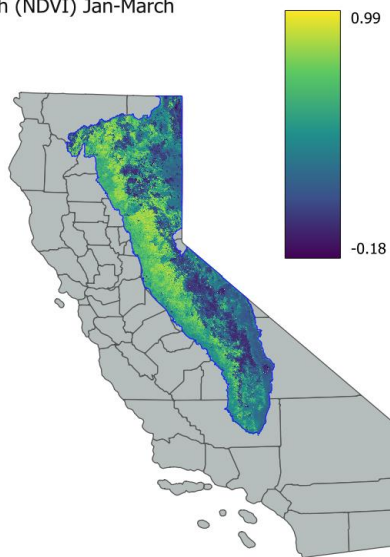
Seasonal NDVI and Climate Patterns (2024) Cont.



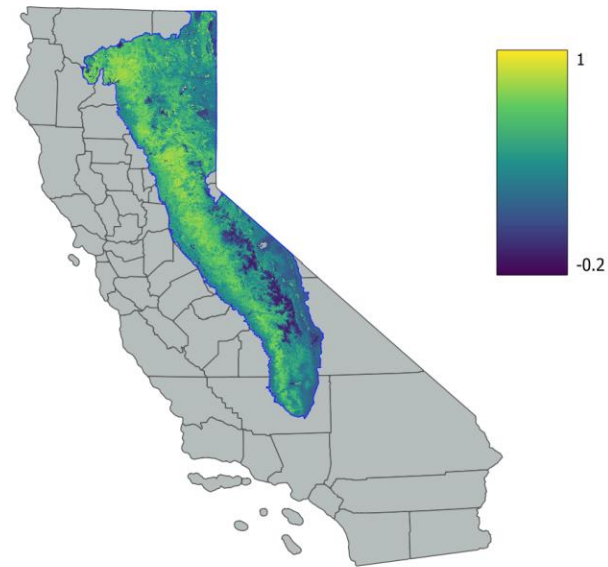


Spatial NDVI Variability and Seasonal Clusters

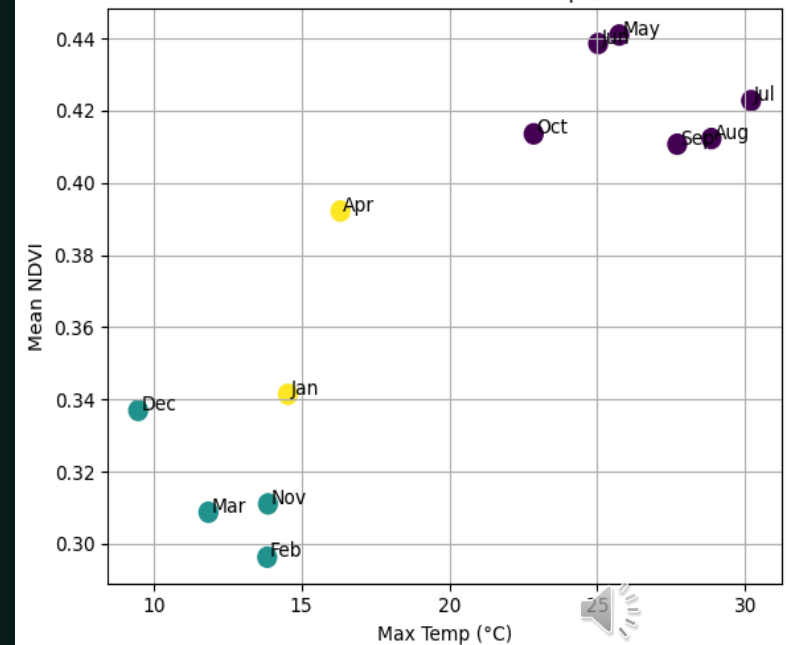
Vegetation Health (NDVI) Jan-March



Vegetation Health June-August



K-Means Clusters: NDVI vs Temp (2024)



Key Findings and Future Work



- NDVI seasonality matches temperature and precipitation cycles.
- Slight upward NDVI trend suggests recovery in 2024.
- Results confirm a Mediterranean-type pattern of growth after rainfall.
- Future extensions: multi-year Cal-Adapt trends, fire data, soil moisture indices



Thank you

